Facial Recognition using AffectNet

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The Problem

- We want to accurately recognize facial expressions in "the wild" (real-world environments). Existing models often fail outside of controlled laboratory settings due to poor lighting, motion blur, occlusions, and significant variations in facial appearance.

SUPRISE

DISGUST

ANGER

HAPPY

SAD

NEUTRAL

CONTEMPT

FEAR

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Related works

- In ***Toward practical smile detection* (Whitehill et al.):** The authors demonstrate that SVMs can be used for smile detection with strong accuracy across varied datasets. They also discuss the value of preprocessing filters such as Gabor energy filters.

- In **Facial Expression Recognition with Occlusions Based on Geometric Representation (Cornejo et al.):** The authors demonstrate both SVM and K-nearest neighbors approaches for expression classification. They additionally implement PCA for dimensionality reduction as a preprocessing step.

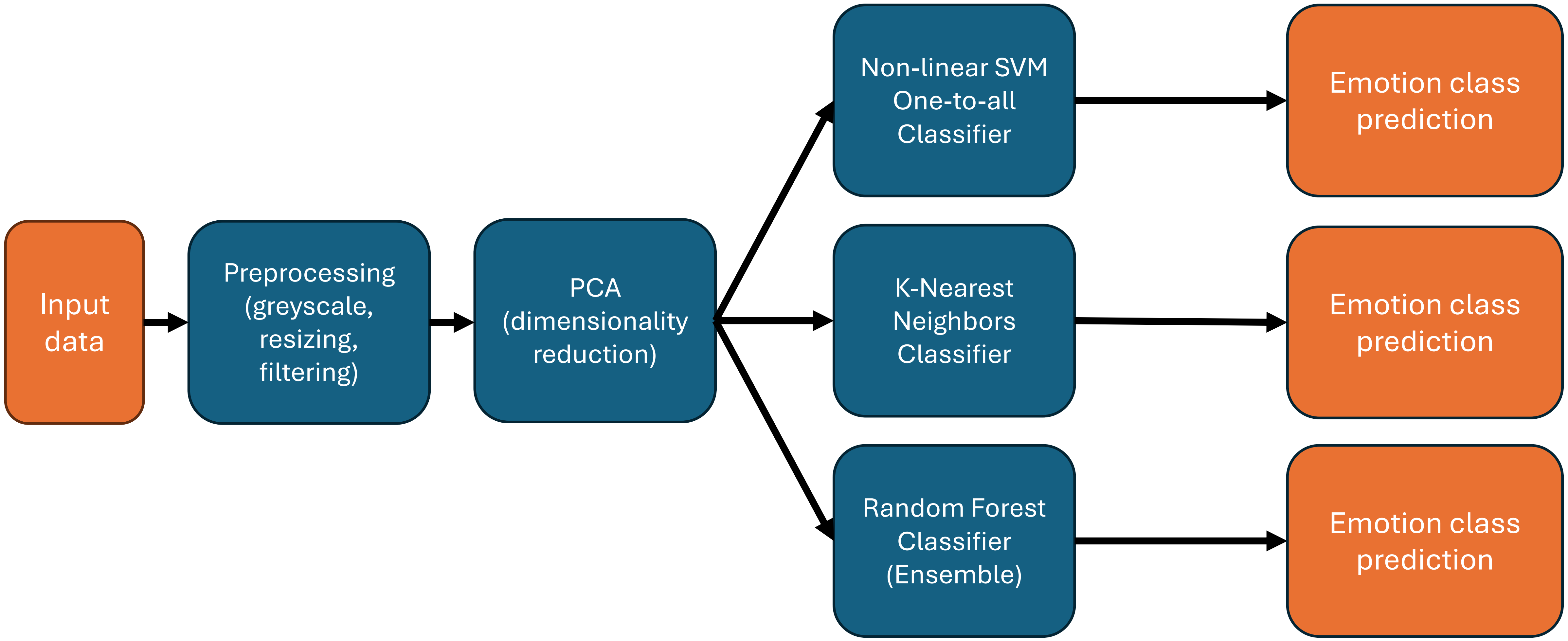
The Data Source and What it looks like

- **Data Retrieval:** We retrieve our data from Kaggle using the `kagglehub` library.
- **Dataset Selection:** Specifically, we are utilizing the **AffectNet** dataset in a YOLO-compatible format.
- **Raw Structure:** Our initial download consists of image files (.jpg, .png) and their corresponding label files (.txt).
- **Storage:** We use our scripts to organize this data into a local project directory under `data/raw/images` and `data/raw/labels`.

Approach

- **Data Ingestion:** We automated the retrieval and organization of the **AffectNet** dataset.
- **Preprocessing:** We implemented quality-gate filtering for blur, exposure, and integrity, followed by a 96 x 96 grayscale standardization.
- **Feature Engineering:** We use principal component analysis on the preprocessed input data to reduce the dimensionality of the feature space
- **Classification Models:**
  - We developed and use two traditional Machine Learning models and one ensemble model to classify the feature reduced input data
  - We built a custom **SVM** implementation which uses Nesterov acceleration and Kernel caching for efficiency.
  - A K-Nearest Neighbors classifier
  - A Random Forest Classifier
- **Evaluation:** We conducted a multi-metric performance analysis evaluating models based on Accuracy, F1 and SMAPE

Classification pipeline



Results: Principal Component Analysis

- Like HW1, our PCA implementation is being used to reduce the dimensionality of greyscale image data.
- We compute the Principal Components via Singular Value Decomposition

Animation showing the reconstruction of an image from the train set while increasing the number of principal components

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Results: Overall Model Performance Statistics

- The overall performance metrics of our three emotion classification models show low accuracy, F1, and SMAPE scores.
- Our SVM performs best in almost all metrics, followed by The KNN and Random Forest methods

Model	Test Accuracy	Test F1 Score	Test SMAPE
SVM	33.00%	0.324	69.88%
KNN	27.12%	0.226	66.31%
Random Forest	24.12%	0.229	78.06%

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Per class results

- per class precision, recall and F-measure are low
- The ‘Happy’ class is an outlier. While the all three models still have weak performance for this class, its performance measures are double the next best performing class

Class	Precision			Recall			F1		
	SVM	KNN	Random Forest	SVM	KNN	Random Forest	SVM	KNN	Random Forest
Anger	0.356	0.800	0.236	0.370	0.040	0.210	0.363	0.076	0.222
Contempt	0.337	0.500	0.123	0.300	0.070	0.090	0.317	0.123	0.104
Disgust	0.242	0.174	0.158	0.230	0.040	0.150	0.236	0.065	0.153
Fear	0.260	0.255	0.296	0.250	0.420	0.240	0.255	0.317	0.265
Happy	0.576	0.439	0.461	0.720	0.800	0.700	0.640	0.567	0.556
Neutral	0.276	0.208	0.155	0.290	0.160	0.170	0.283	0.181	0.162
Sad	0.270	0.294	0.237	0.270	0.200	0.230	0.270	0.238	0.234
Surprise	0.244	0.165	0.136	0.210	0.440	0.140	0.226	0.240	0.138

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Conclusion

- From our testing, the **SVM**, **KNN**, and **Random Forest** models that we developed are all unable to effectively discriminate between the different emotion classes in the AffectNet Dataset.
- While the works that we based our approach on (Whitehill et al., Cornejo et al.) use similar techniques to us, they focus on different datasets
  - Whitehill et al. merely focuses on Smile Detection rather than multi-class emotion recognition
- Further research into techniques that have been effective with the AffectNet dataset should be done in order to improve results

Future Work

To make this research accessible for future developers, we can build upon our foundation in the following ways:

- **Using smarter image scanners :** We can swap PCA for models like ResNet to better detect tiny, complex "micro-expressions" that standard methods often miss.
- **Add an Intelligent Manager:** We can use a "meta-learner" to automatically pick the best model—SVM, KNN, or Random Forest—for each specific emotion.
- **Improve Transparency:** We can use SHAP to create heatmaps that show users exactly which facial features, like a smile or a frown, caused the model to make its decision.

Future Work

There are several areas that could be explored to improve the classification performance of our models

- **Gabor Filtering :** *Whitehill et al*. and *Cornejo et al*. use gabor energy filters as a stage in their preprocessing pipelines
- **Reduced # of output classes:** reducing the number of emotions that we train our model to discriminate between could improve performance.
- **Using a one-to-one SVM:** Changing our SVM classifier to base decision on more fine grain binary classifier could help discriminate between emotions.

References

- Data source: https://www.mohammadmahoor.com/pages/databases/affectnetplus/
- Fard, A. P., Hosseini, M. M., Sweeny, T. D., & Mahoor, M. H. (2024). AffectNet+: A Database for Enhancing Facial Expression Recognition with Soft-Labels. arXiv preprint arXiv:2410.22506.
- J. Whitehill, G. Littlewort, I. Fasel, M. Bartlett and J. Movellan, "Toward Practical Smile Detection," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 31, no. 11, pp. 2106-2111, Nov. 2009, https://doi.org/10.1109/TPAMI.2009.42
- Cornejo, J.Y.R., Pedrini, H., Flórez-Revuelta, F. (2015). Facial Expression Recognition with Occlusions Based on Geometric Representation. In: Pardo, A., Kittler, J. (eds) Progress in Pattern Recognition, Image Analysis, Computer Vision, and Applications. CIARP 2015. Lecture Notes in Computer Science, vol 9423. Springer, Cham. https://doi.org/10.1007/978-3-319-25751-8_32